

Efficiency and resource measurement

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dataset shift in neural models

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compare under requirements for Efficiency and resource measurement. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Efficiency and resource measurement. Proposed explanation: “Fewer parameters always means faster”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Evaluate a new case independently

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time. Give median times, amortised times, throughput, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per output: find its parameter count and raw bytes at four bytes each.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Summarise repeated timings

For A and B, calculate median pass time, amortised milliseconds per example and median-pass throughput. State the measured boundary and one latency claim these numbers cannot support.

My reasoning and evidence

6. W2 Compare under requirements

With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage throughput increase differs.

My reasoning and evidence

7. W3 Separate storage and memory

Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak training memory or an energy measurement.

My reasoning and evidence

8. W4 Evaluate a new case independently

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time. Give median times, amortised times, throughput, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per output: find its parameter count and raw bytes at four bytes each.

My reasoning and evidence

9. W5 Repair an unfair benchmark

A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning training run are reported. Explain how to compare and report the results fairly.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled efficiency and resource measurement evidence sheet to Image-pattern Network Investigation. Show how you can calculate throughput and amortised time per example from repeated timings. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Efficiency and resource measurement

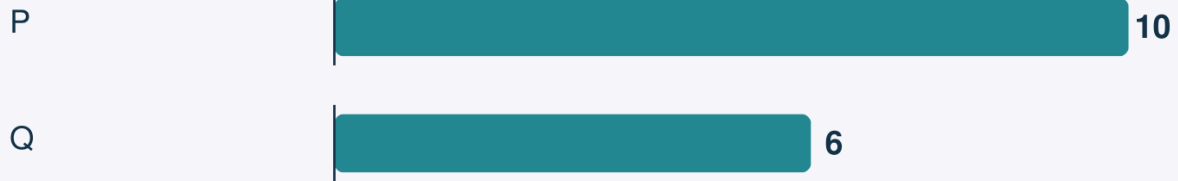
Compare speed and quality under the same case count and acceptance contract.

Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Require accuracy $\geq 93\%$ and mean amortised time $\leq 12\text{ms/case}$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Mean amortised time, not single-request latency

ms per case



Bar length shows the supplied value; the marked baseline is zero.

Efficiency and resource measurement

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Summarise P

Mean time 0.40s; 10ms/case; accuracy 38/40=95%:
passes both.



2. Summarise Q

Mean time 0.24s; 6ms/case; accuracy 36/40=90%:
fails quality.



3. Choose eligible candidate

P meets the declared contract; Q is faster but ineligible.

CHANGE ONE THING

Lower the accuracy requirement to 90% before a new selection exercise.

Explain the changed result using the matching lesson.

Why is 10ms/case not necessarily a single-request latency? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare speed and quality under the same case count and acceptance contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? P meets the declared contract; Q is faster but ineligible.

1. Summarise P
2. Summarise Q
3. Choose eligible candidate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Choose eligible candidate P meets the declared contract; Q is faster but ineligible. Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate. Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Practice 2: explain the deciding evidence

Changed-case practice: Lower the accuracy requirement to 90% before a new selection exercise. Which conclusion is supported by the supplied facts?

1. Both become eligible; Q has the smaller measured mean amortised time.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q's measured accuracy did not improve.

Repair a missing building block

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed input mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted output head. Short practice: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
